COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	29/10/2026	**DATE SUBMITTED:**	29/10/2026
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Practical Exam	**Final Hands-on Skills Exam**

Github Link:https://github.com/DamienJakeGestiada/CPE212_JakeGestiada/tree/school-lab/Final_Exam_Gestiada

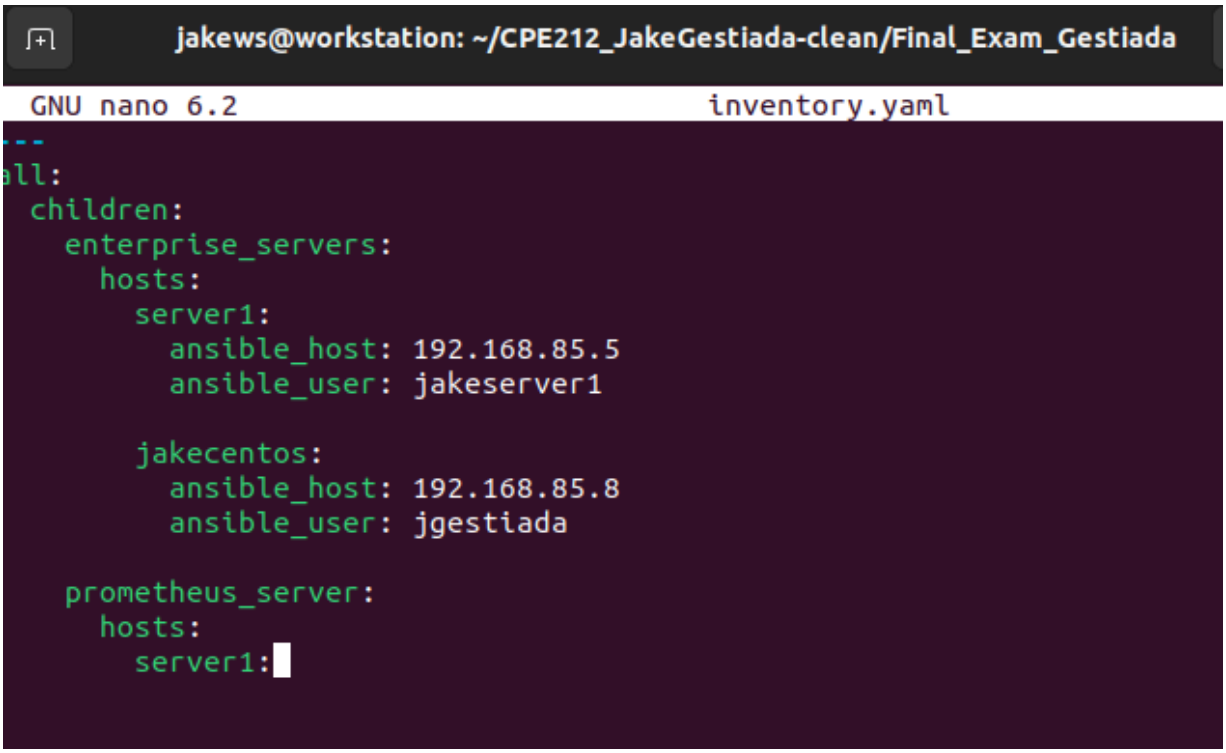
Tools Needed:

1. VM with Ubuntu, CentOS and Ansible installed (Control node = jakews create a playbook, manage node 1 ubuntu and 1 centos)

2. Web browser

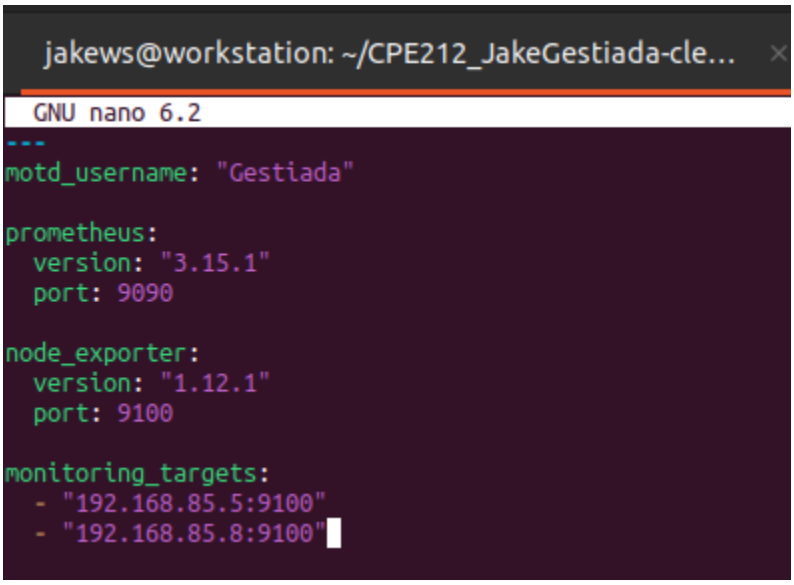
Procedure:

1. Create a folder add to repository and label it as "Final_Exam_Gestiada"



- This is a connection from the servers centos and ubuntu and prometheus web server.

2. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

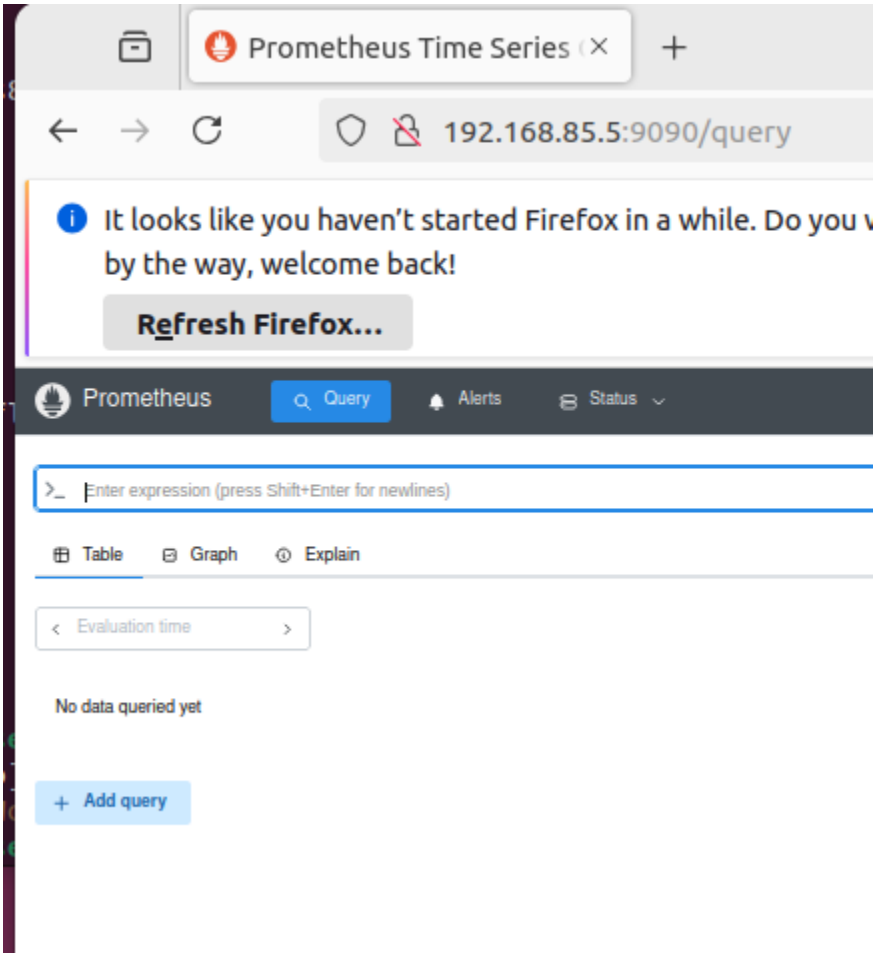


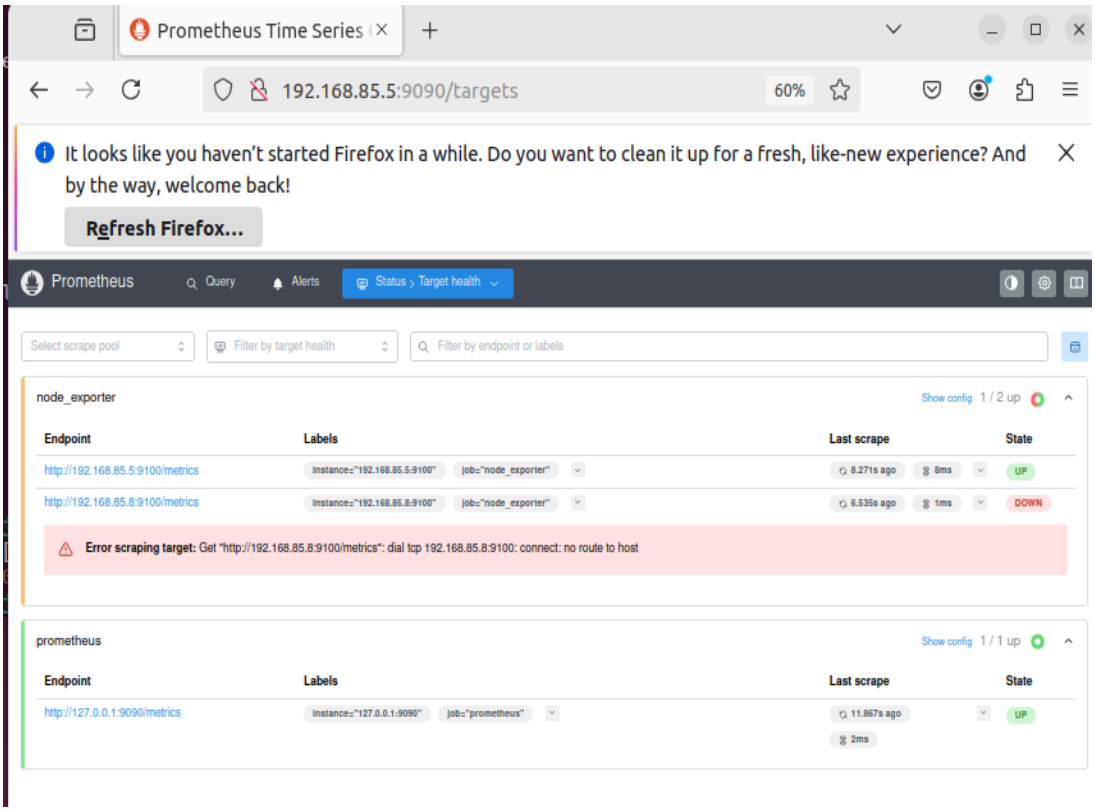
```
jakews@workstation:~/CPE212_JakeGestiada-clean/Final_Exam_Gestiada$ ls
config.yaml  final_exam.yaml  inventory.yaml
```

- This code is built to contain the settings needed for our playbook.This include the MOTD message, Prometheus and Node Exporter versions, and the ports used for monitoring. It also includes the IP addresses of our Ubuntu and CentOS servers so Prometheus knows which machines to monitor.

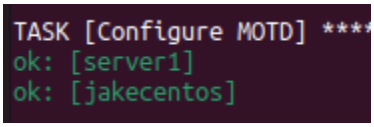
  3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

  3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)





4.4 Change Motd as "Ansible Managed by Gestiada"



4. Push and commit your files in GitHub

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

```
jakews@workstation:~/CPE212_JakeGestiada-clean/Final_Exam_Gestiada$ ansible-playbook -i inventory.yaml final_exam.yaml -K
BECOME password:

PLAY [Configure MOTD and Node Exporter] ****************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host jakecentos should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using
the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature
will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in
ansible.cfg.
ok: [jakecentos]

TASK [Configure MOTD] **********************************************************************************************
ok: [server1]
ok: [jakecentos]

TASK [Create Node Exporter system user] ****************************************************************************
ok: [server1]
ok: [jakecentos]

TASK [Download Node Exporter] **************************************************************************************
ok: [jakecentos]
fatal: [server1]: FAILED! => {"changed": false, "dest": "/tmp/node_exporter.tar.gz", "elapsed": 15, "msg": "Request failed
: <urlopen error [Errno -3] Temporary failure in name resolution>", "url": "https://github.com/prometheus/node_exporter/re
leases/download/v1.12.1/node_exporter-1.12.1.linux-amd64.tar.gz"}

TASK [Create Node Exporter extraction directory] *******************************************************************
changed: [jakecentos]

TASK [Extract Node Exporter] ***************************************************************************************
changed: [jakecentos]
```

```
TASK [Download Node Exporter] **************************************************************************************
ok: [jakecentos]
fatal: [server1]: FAILED! => {"changed": false, "dest": "/tmp/node_exporter.tar.gz", "elapsed": 15, "msg": "Request failed
: <urlopen error [Errno -3] Temporary failure in name resolution>", "url": "https://github.com/prometheus/node_exporter/re
leases/download/v1.12.1/node_exporter-1.12.1.linux-amd64.tar.gz"}

TASK [Create Node Exporter extraction directory] *******************************************************************
changed: [jakecentos]

TASK [Extract Node Exporter] ***************************************************************************************
changed: [jakecentos]

TASK [Install Node Exporter binary] ********************************************************************************
ok: [jakecentos]

TASK [Configure Node Exporter service] *****************************************************************************
changed: [jakecentos]

TASK [Start and enable Node Exporter] ******************************************************************************
ok: [jakecentos]

RUNNING HANDLER [Restart Node Exporter] ****************************************************************************
changed: [jakecentos]

PLAY [Install and Configure Prometheus] ****************************************************************************

PLAY RECAP *********************************************************************************************************
jakecentos                 : ok=10   changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=3    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

jakews@workstation: ~/CPE212_JakeGestiada-cle... × jakews@workstation: ~/CPE212_JakeGestia

GNU nano 6.2 final_exam.yaml

```yaml
---

- name: Download Monitoring Archives on Control Node
  hosts: localhost
  connection: local
  gather_facts: false

  vars_files:
    - config.yaml
```

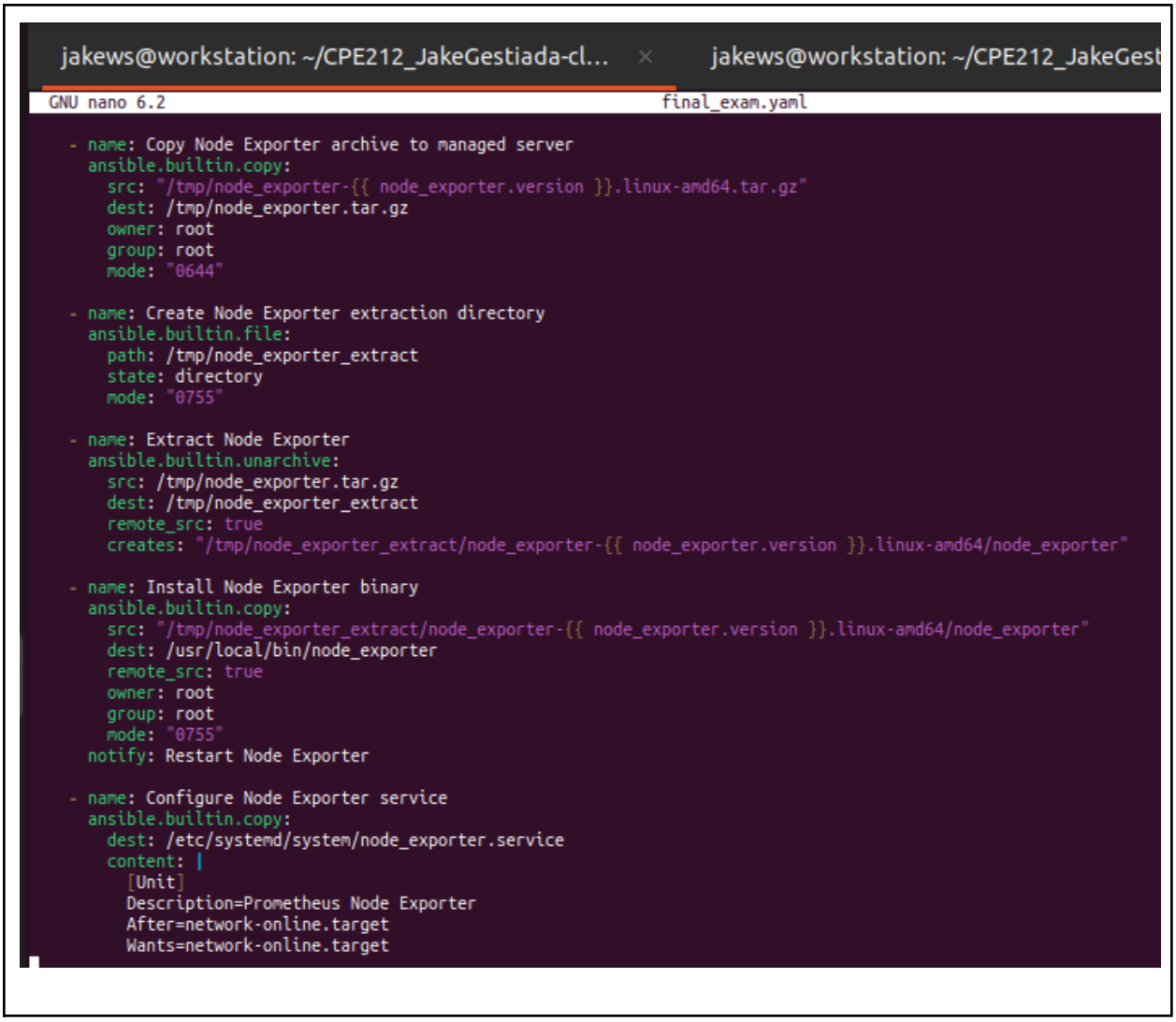
```yaml
- name: Configure MOTD and Node Exporter
  hosts: enterprise_servers
  become: true
  gather_facts: true

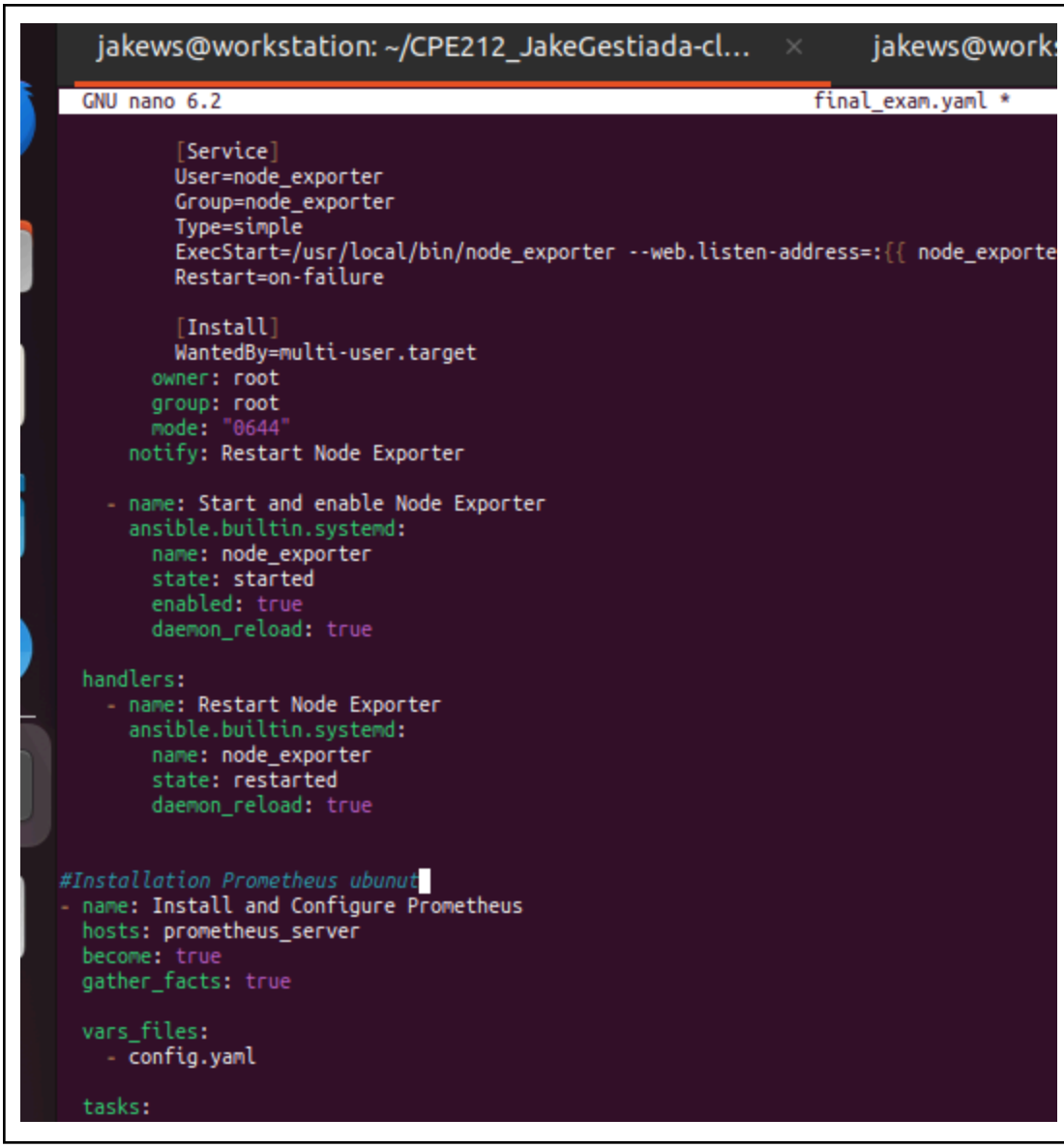
  vars_files:
    - config.yaml

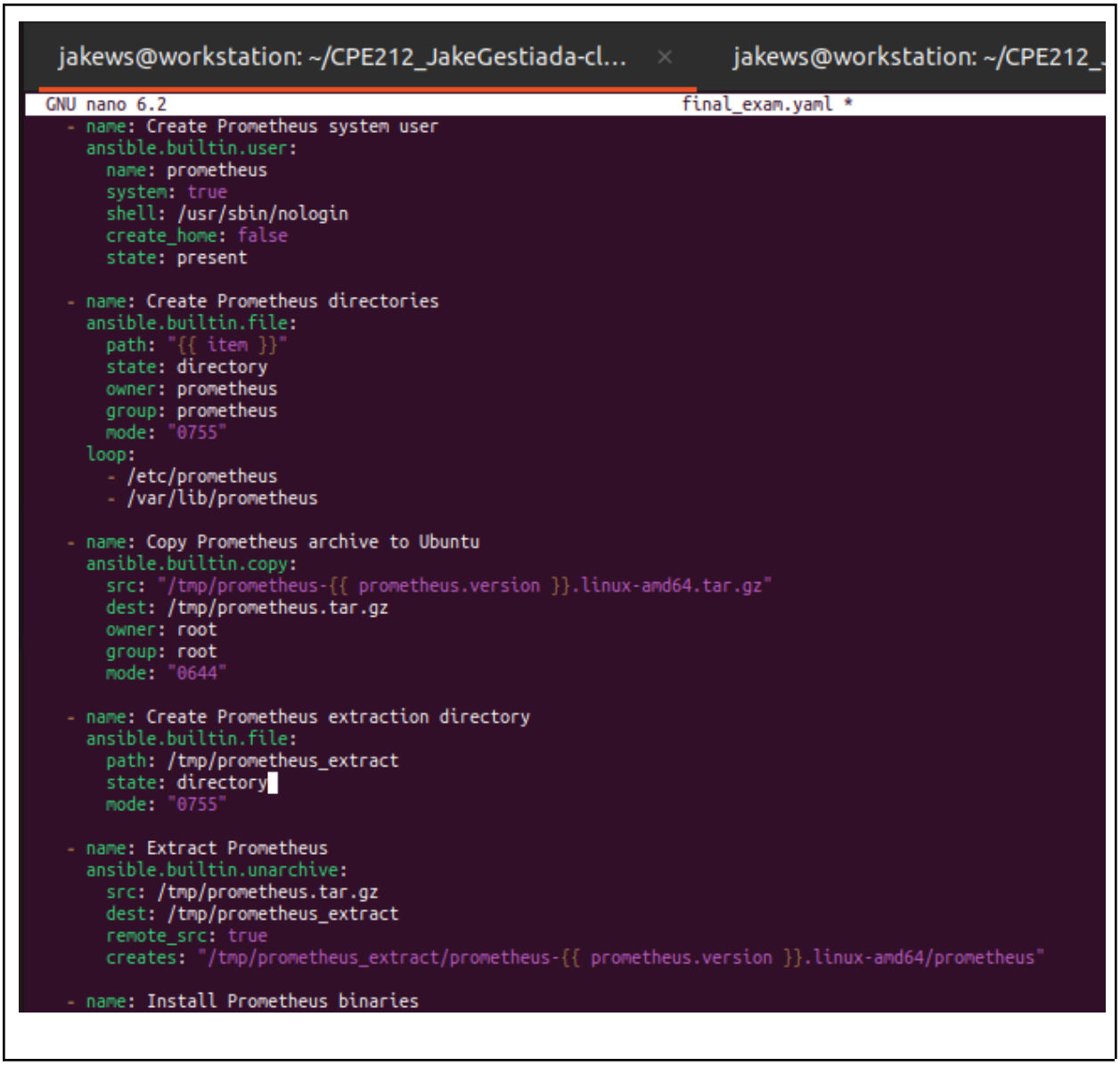
  tasks:
    - name: Configure MOTD
      ansible.builtin.copy:
        dest: /etc/motd
        content: "Ansible Managed by {{ motd_username }}\n"
        owner: root
        group: root
        mode: "0644"

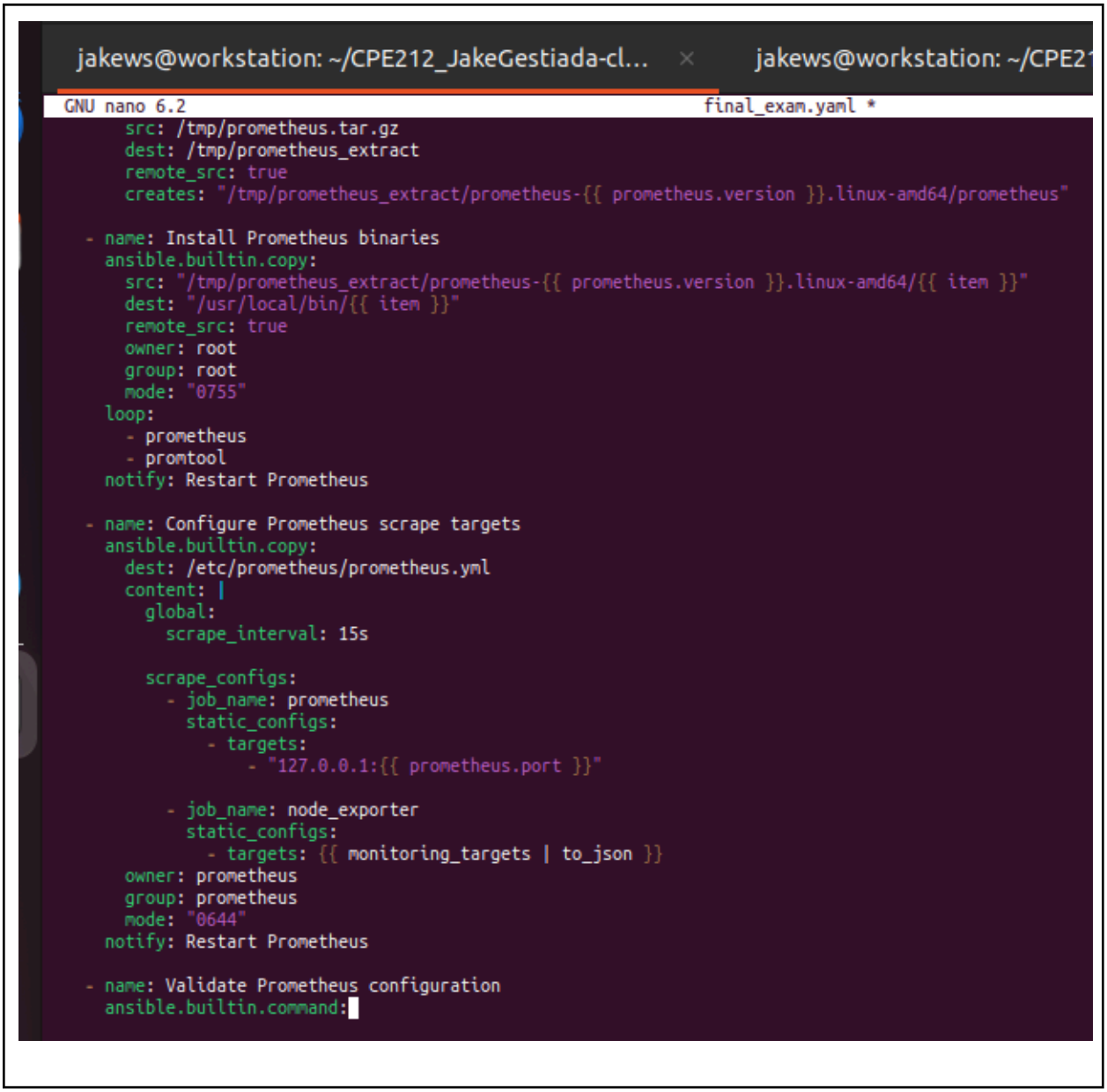
    - name: Create Node Exporter system user
      ansible.builtin.user:
        name: node_exporter
        system: true
        shell: /usr/sbin/nologin
        create_home: false
        state: present

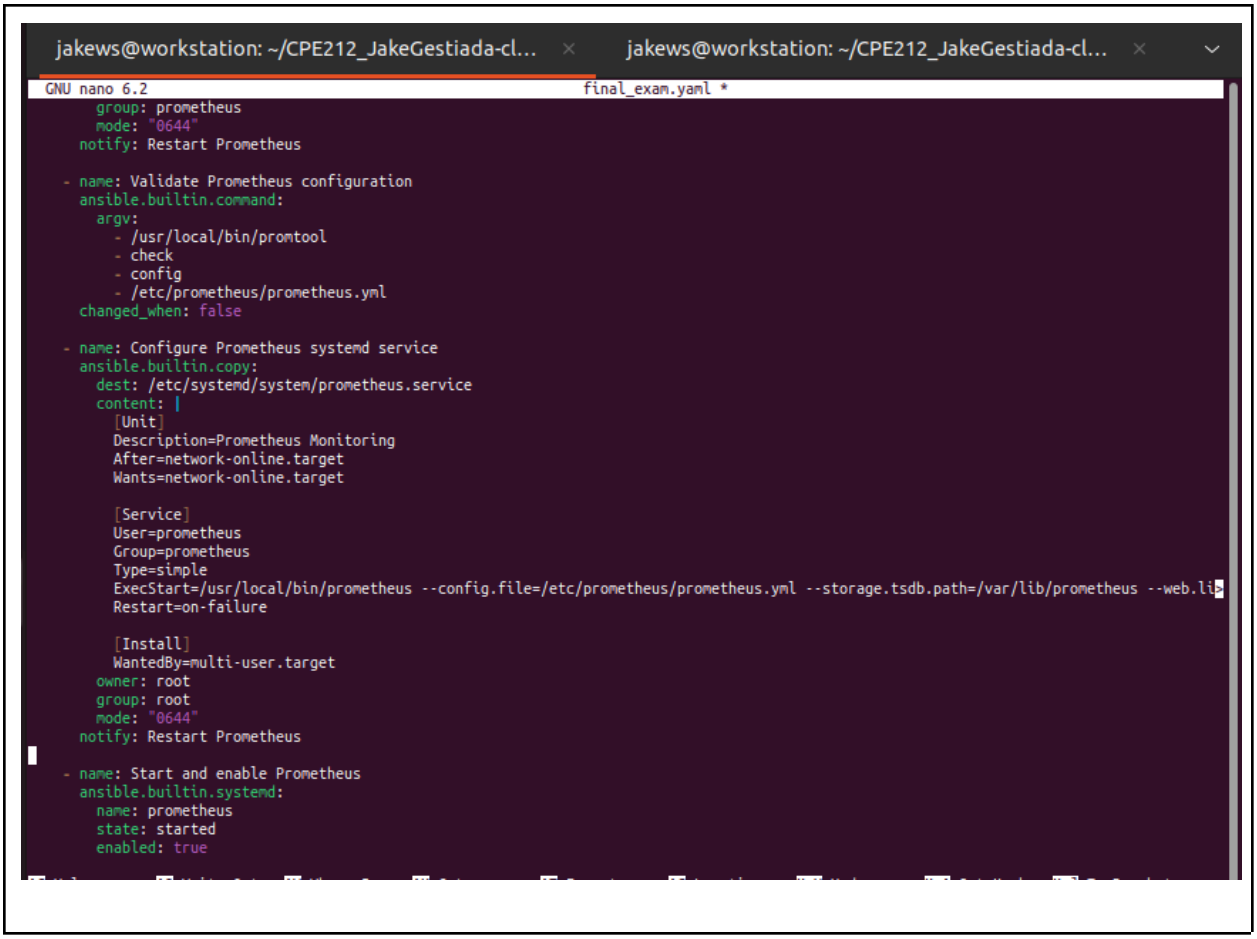
    - name: Copy Node Exporter archive to managed server
      ansible.builtin.copy:
        src: "/tmp/node_exporter-{{ node_exporter.version }}.linux-amd64.tar.gz"
        dest: /tmp/node_exporter.tar.gz
        owner: root
        group: root
```

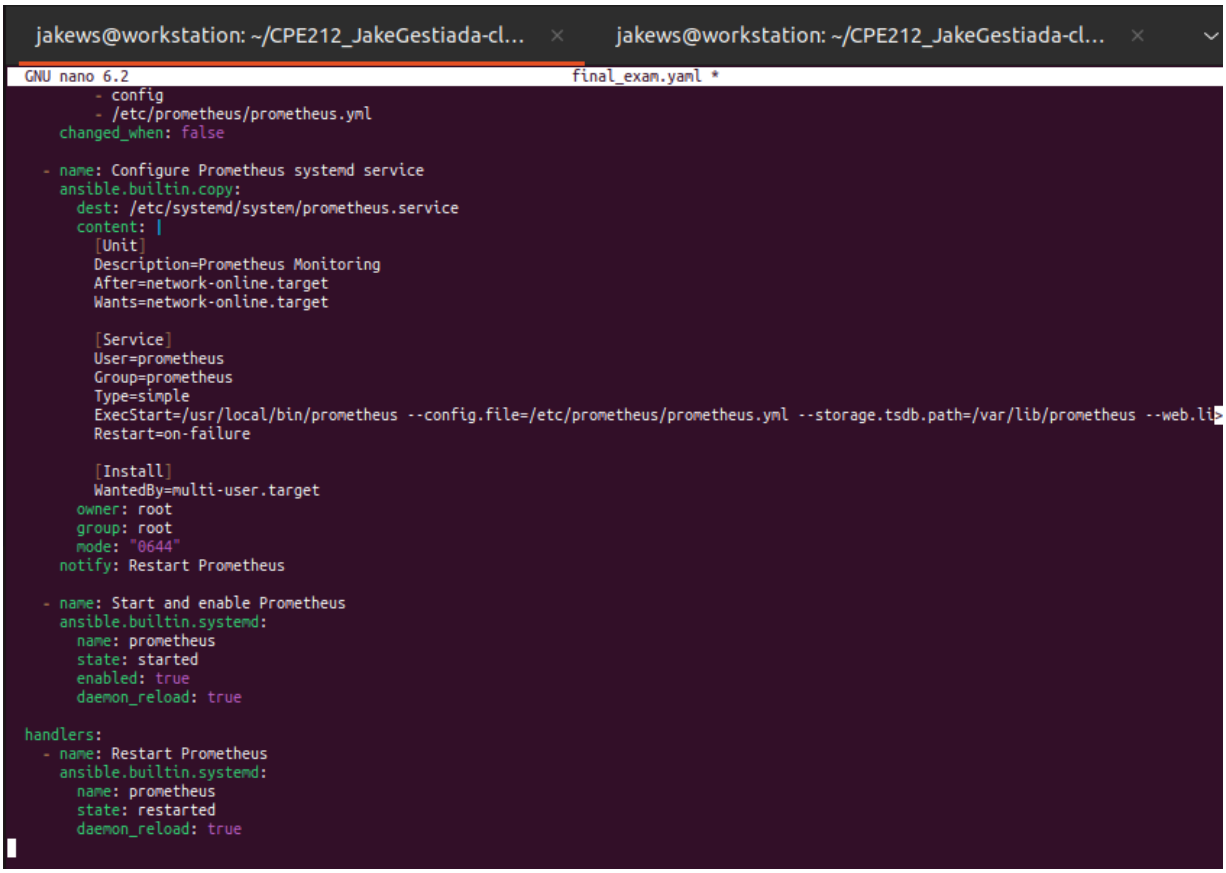

jakews@workstation: ~/CPE212_JakeGestiada-cl...
jakews@workstation: ~/CPE212_JakeGest
GNU nano 6.2
final_exam.yaml
- name: Copy Node Exporter archive to managed server
ansible.builtin.copy:
src: "/tmp/node_exporter-{{ node_exporter.version }}.linux-amd64.tar.gz"
dest: /tmp/node_exporter.tar.gz
owner: root
group: root
mode: "0644"
- name: Create Node Exporter extraction directory
ansible.builtin.file:
path: /tmp/node_exporter_extract
state: directory
mode: "0755"
- name: Extract Node Exporter
ansible.builtin.unarchive:
src: /tmp/node_exporter.tar.gz
dest: /tmp/node_exporter_extract
remote_src: true
creates: "/tmp/node_exporter_extract/node_exporter-{{ node_exporter.version }}.linux-amd64/node_exporter"
- name: Install Node Exporter binary
ansible.builtin.copy:
src: "/tmp/node_exporter_extract/node_exporter-{{ node_exporter.version }}.linux-amd64/node_exporter"
dest: /usr/local/bin/node_exporter
remote_src: true
owner: root
group: root
mode: "0755"
notify: Restart Node Exporter
- name: Configure Node Exporter service
ansible.builtin.copy:
dest: /etc/systemd/system/node_exporter.service
content: |
[Unit]
Description=Prometheus Node Exporter
After=network-online.target
Wants=network-online.target


jakews@workstation: ~/CPE212_JakeGestiada-cl...
jakews@works
GNU nano 6.2
final_exam.yaml *
[Service]
User=node_exporter
Group=node_exporter
Type=simple
ExecStart=/usr/local/bin/node_exporter --web.listen-address=:{{ node_exporte
Restart=on-failure
[Install]
WantedBy=multi-user.target
owner: root
group: root
mode: "0644"
notify: Restart Node Exporter
- name: Start and enable Node Exporter
ansible.builtin.systemd:
name: node_exporter
state: started
enabled: true
daemon_reload: true
handlers:
- name: Restart Node Exporter
ansible.builtin.systemd:
name: node_exporter
state: restarted
daemon_reload: true
#Installation Prometheus ubunut
- name: Install and Configure Prometheus
hosts: prometheus_server
become: true
gather_facts: true
vars_files:
- config.yaml
tasks:


jakews@workstation: ~/CPE212_JakeGestiada-cl...
jakews@workstation: ~/CPE212_
GNU nano 6.2
final_exam.yaml *
- name: Create Prometheus system user
ansible.builtin.user:
name: prometheus
system: true
shell: /usr/sbin/nologin
create_home: false
state: present
- name: Create Prometheus directories
ansible.builtin.file:
path: "{{ item }}"
state: directory
owner: prometheus
group: prometheus
mode: "0755"
loop:
- /etc/prometheus
- /var/lib/prometheus
- name: Copy Prometheus archive to Ubuntu
ansible.builtin.copy:
src: "/tmp/prometheus-{{ prometheus.version }}.linux-amd64.tar.gz"
dest: /tmp/prometheus.tar.gz
owner: root
group: root
mode: "0644"
- name: Create Prometheus extraction directory
ansible.builtin.file:
path: /tmp/prometheus_extract
state: directory
mode: "0755"
- name: Extract Prometheus
ansible.builtin.unarchive:
src: /tmp/prometheus.tar.gz
dest: /tmp/prometheus_extract
remote_src: true
creates: "/tmp/prometheus_extract/prometheus-{{ prometheus.version }}.linux-amd64/prometheus"
- name: Install Prometheus binaries


jakews@workstation: ~/CPE212_JakeGestiada-cl...
jakews@workstation: ~/CPE2
GNU nano 6.2
final_exam.yaml *
src: /tmp/prometheus.tar.gz
dest: /tmp/prometheus_extract
remote_src: true
creates: "/tmp/prometheus_extract/prometheus-{{ prometheus.version }}.linux-amd64/prometheus"
- name: Install Prometheus binaries
ansible.builtin.copy:
src: "/tmp/prometheus_extract/prometheus-{{ prometheus.version }}.linux-amd64/{{ item }}"
dest: "/usr/local/bin/{{ item }}"
remote_src: true
owner: root
group: root
mode: "0755"
loop:
- prometheus
- promtool
notify: Restart Prometheus
- name: Configure Prometheus scrape targets
ansible.builtin.copy:
dest: /etc/prometheus/prometheus.yml
content: |
global:
scrape_interval: 15s
scrape_configs:
- job_name: prometheus
static_configs:
- targets:
- "127.0.0.1:{{ prometheus.port }}"
- job_name: node_exporter
static_configs:
- targets: {{ monitoring_targets | to_json }}
owner: prometheus
group: prometheus
mode: "0644"
notify: Restart Prometheus
- name: Validate Prometheus configuration
ansible.builtin.command:


jakews@workstation: ~/CPE212_JakeGestiada-cl...
jakews@workstation: ~/CPE212_JakeGestiada-cl...
GNU nano 6.2
final_exam.yaml *
group: prometheus
mode: "0644"
notify: Restart Prometheus
- name: Validate Prometheus configuration
ansible.builtin.command:
argv:
- /usr/local/bin/promtool
- check
- config
- /etc/prometheus/prometheus.yml
changed_when: false
- name: Configure Prometheus systemd service
ansible.builtin.copy:
dest: /etc/systemd/system/prometheus.service
content: |
[Unit]
Description=Prometheus Monitoring
After=network-online.target
Wants=network-online.target
[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus --web.li
Restart=on-failure
[Install]
WantedBy=multi-user.target
owner: root
group: root
mode: "0644"
notify: Restart Prometheus
- name: Start and enable Prometheus
ansible.builtin.systemd:
name: prometheus
state: started
enabled: true



- The playbook automates the installation and configuration of Prometheus and Node Exporter on the Ubuntu and CentOS servers. Prometheus uses the code from the previous activities to collect and monitor server metrics, while Node Exporter is installed using the download link from the website I searched. The playbook also configures the MOTD to display “Ansible Managed by Gestiada” on both servers.

Prometheus link on server 1

http://192.168.85.5:9090

http://192.168.85.5:9090/targets

5. For your final exam to be counted, please paste your repository link as an answer in this exam.

Note: Extra points if you will implement the said services via containerization.